COHOMOLOGICAL DIMENSION D CONTROLS BASE CHANGE OF THE TOP DIRECT IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. Verbatim author omission in the counted span: 'Any right exact R'-linear functor F which commutes with direct sums is given by tensoring with an R'-module (omitted; left as exercise for the reader).' This is the named Eilenberg-Watts theorem, an external established result; the load-bearing citation is to a classic, and (a)/(c) are fully worked in span. Flagged needs_human_spotcheck for exactly this sentence.

Allowed standard prerequisites (editorial).

- Eilenberg-Watts tensor representation. classic theorem; the source marks the step '(omitted; left as exercise for the reader)'

Source reference. Original reference 08HY: sources/stacks-chow/tags.txt, source lines 10796–10796. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01PE: sources/stacks-chow/tags.txt, source lines 2227–2227. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01I9: sources/stacks-chow/tags.txt, source lines 1970–1970. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XK: sources/stacks-chow/tags.txt, source lines 2521–2521. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01FF: sources/stacks-chow/tags.txt, source lines 1868–1868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XJ: sources/stacks-chow/tags.txt, source lines 2520–2520. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $d \geq 0$. Assume*

(1) *X and Y are quasi-compact and quasi-separated, and*
(2) *$R^i f_* \mathcal{F} = 0$ for $i > d$ and every quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$.*

Then we have

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(a) *for any base change diagram*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

we have $R^i f'_* \mathcal{F}' = 0$ for $i > d$ and any quasi-coherent $\mathcal{O}_{X'}$-module $\mathcal{F}'$,

(b) *$R^d f'_*(\mathcal{F}' \otimes_{\mathcal{O}_{X'}} (f')^* \mathcal{G}') = R^d f'_* \mathcal{F}' \otimes_{\mathcal{O}_{Y'}} \mathcal{G}'$ for any quasi-coherent $\mathcal{O}_{Y'}$-module $\mathcal{G}'$,*

(c) *formation of $R^d f'_* \mathcal{F}'$ commutes with arbitrary further base change (see proof for explanation).*

Proof. Before giving the proofs, we explain the meaning of (c). Suppose we have an additional cartesian square

$$\begin{array}{ccccc} X'' & \xrightarrow{h'} & X' & \xrightarrow{g'} & X \\ {\scriptstyle f''}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y'' & \xrightarrow{h} & Y' & \xrightarrow{g} & Y \end{array}$$

tacked onto our given diagram. If (a) holds, then there is a canonical map $\gamma : h^* R^d f'_* \mathcal{F}' \to R^d f''_*(h')^* \mathcal{F}'$. Namely, γ is the map on degree d cohomology sheaves induced by the composition

$$Lh^* Rf'_* \mathcal{F}' \longrightarrow Rf''_* L(h')^* \mathcal{F}' \longrightarrow Rf''_*(h')^* \mathcal{F}'$$

Here the first arrow is the base change map (Cohomology, Remark 08HY) and the second arrow complex from the canonical map $L(g')^* \mathcal{F} \to (g')^* \mathcal{F}$. Similarly, since $Rf'_* \mathcal{F}$ has no nonzero cohomology sheaves in degrees $> d$ by (a) we have $H^d(Lh^* Rf_* \mathcal{F}') = h^* R^d f_* \mathcal{F}$. The content of (c) is that γ is an isomorphism.

Having said this, we can check (a), (b), and (c) locally on Y' and Y''. Suppose that $V \subset Y$ is a quasi-compact open subscheme. Then we claim (1) and (2) hold for $f|_{f^{-1}(V)} : f^{-1}(V) \to V$. Namely, (1) is immediate and (2) follows because any quasi-coherent module on $f^{-1}(V)$ is the restriction of a quasi-coherent module on X (Properties, Lemma 01PE) and formation of higher direct images commutes with restriction to opens. Thus we may also work locally on Y. In other words, we may assume Y'', Y', and Y are affine schemes.

Proof of (a) when Y' and Y are affine. In this case the morphisms g and g' are affine. Thus $g_* = Rg_*$ and $g'_* = Rg'_*$ (Cohomology of Schemes, Lemma 01XC) and g_* is identified with the restriction functor on modules (Schemes, Lemma 01I9). Then

$$g_*(R^i f'_* \mathcal{F}') = H^i(Rg_* Rf'_* \mathcal{F}') = H^i(Rf_* Rg'_* \mathcal{F}') = H^i(Rf_* g'_* \mathcal{F}') = Rf^i_* g'_* \mathcal{F}'$$

which is zero by assumption (2). Hence (a) by our description of g_*.

Proof of (b) when Y' is affine, say $Y' = \mathrm{Spec}(R')$. By part (a) we have $H^{d+1}(X', \mathcal{F}') = 0$ for any quasi-coherent $\mathcal{O}_{X'}$-module $\mathcal{F}'$, see Cohomology of Schemes, Lemma 01XK. Consider the functor F on R'-modules defined by the rule

$$F(M) = H^d(X', \mathcal{F}' \otimes_{\mathcal{O}_{X'}} (f')^* \widetilde{M})$$

By Cohomology, Lemma 01FF this functor commutes with direct sums (this is where we use that X and hence X' is quasi-compact and quasi-separated). On the other hand, if $M_1 \to M_2 \to M_3 \to 0$ is an exact sequence, then

$$\mathcal{F}' \otimes_{\mathcal{O}_{X'}} (f')^* \widetilde{M}_1 \to \mathcal{F}' \otimes_{\mathcal{O}_{X'}} (f')^* \widetilde{M}_2 \to \mathcal{F}' \otimes_{\mathcal{O}_{X'}} (f')^* \widetilde{M}_3 \to 0$$

is an exact sequence of quasi-coherent modules on X' and by the vanishing of higher cohomology given above we get an exact sequence

$$F(M_1) \to F(M_2) \to F(M_3) \to 0$$

In other words, F is right exact. Any right exact R'-linear functor $F : \mathrm{Mod}_{R'} \to \mathrm{Mod}_{R'}$ which commutes with direct sums is given by tensoring with an R'-module (omitted; left as exercise for the reader). Thus we obtain $F(M) = H^d(X', \mathcal{F}') \otimes_{R'} M$. Since $R^d(f')_*\mathcal{F}'$ and $R^d(f')_*(\mathcal{F}' \otimes_{\mathcal{O}_{X'}} (f')^*\widetilde{M})$ are quasi-coherent (Cohomology of Schemes, Lemma 01XJ), the fact that $F(M) = H^d(X', \mathcal{F}') \otimes_{R'} M$ translates into the statement given in (b).

Proof of (c) when $Y'' \to Y' \to Y$ are morphisms of affine schemes. Say $Y'' = \mathrm{Spec}(R'')$ and $Y' = \mathrm{Spec}(R')$. Then we see that $R^d f''_*(h')^*\mathcal{F}'$ is the quasi-coherent module on Y' associated to the R''-module $H^d(X'', (h')^*\mathcal{F}')$. Now $h' : X'' \to X'$ is affine hence $H^d(X'', (h')^*\mathcal{F}') = H^d(X, h'_*(h')^*\mathcal{F}')$ by the already used Cohomology of Schemes, Lemma 089W. We have

$$h'_*(h')^*\mathcal{F}' = \mathcal{F}' \otimes_{\mathcal{O}_{X'}} (f')^*\widetilde{R''}$$

as the reader sees by checking on an affine open covering. Thus $H^d(X'', (h')^*\mathcal{F}') = H^d(X', \mathcal{F}') \otimes_{R'} R''$ by part (b) applied to f' and the proof is complete. □